

Boundary Value Problem

In many engineering problems, you will have to solve ordinary differential equations that look something like

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = r(x) \quad \text{—————} \quad (1)$$

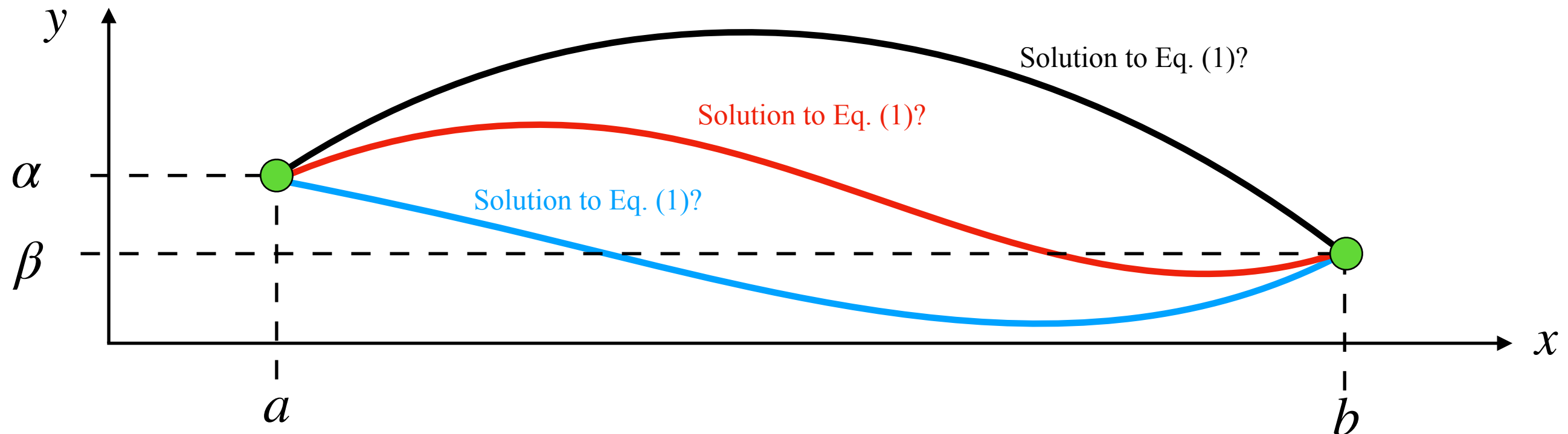
where

$$a \leq x \leq b$$

We are also usually given the conditions at the boundaries

$$\left\{ \begin{array}{l} y(a) = \alpha \\ y(b) = \beta \end{array} \right\}$$

Values of y given at the boundaries:
Dirichlet boundary conditions.



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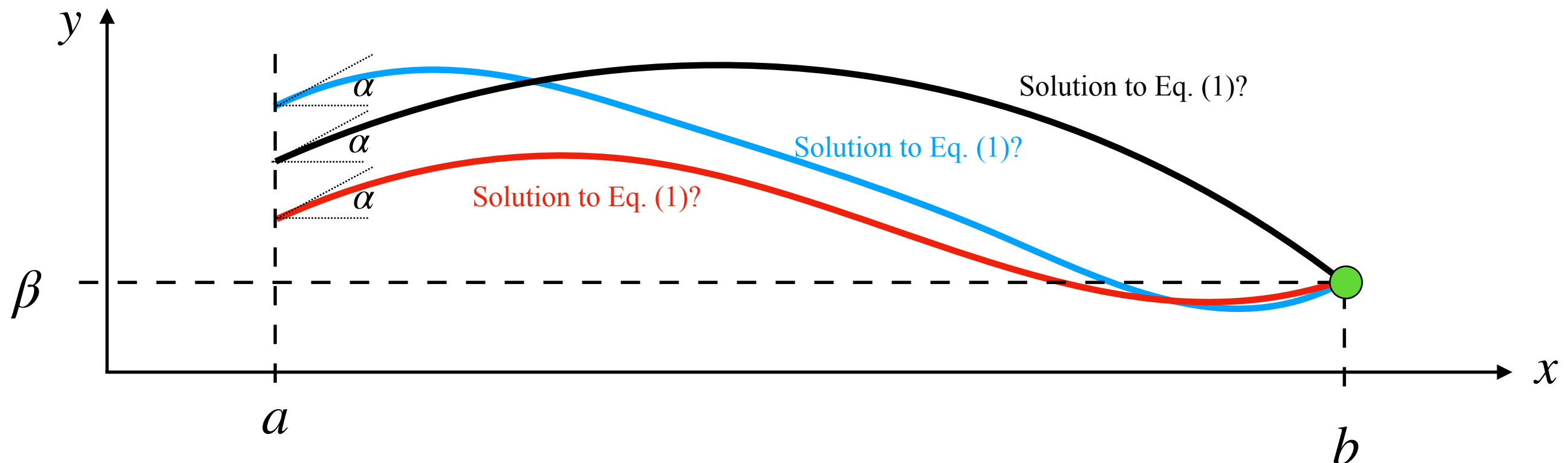
We are also usually given the conditions at the boundaries

$$\left\{ \begin{array}{l} y'(a) = \alpha \\ y(b) = \beta \end{array} \right.$$

Value of derivative
y given at the $x = a$:

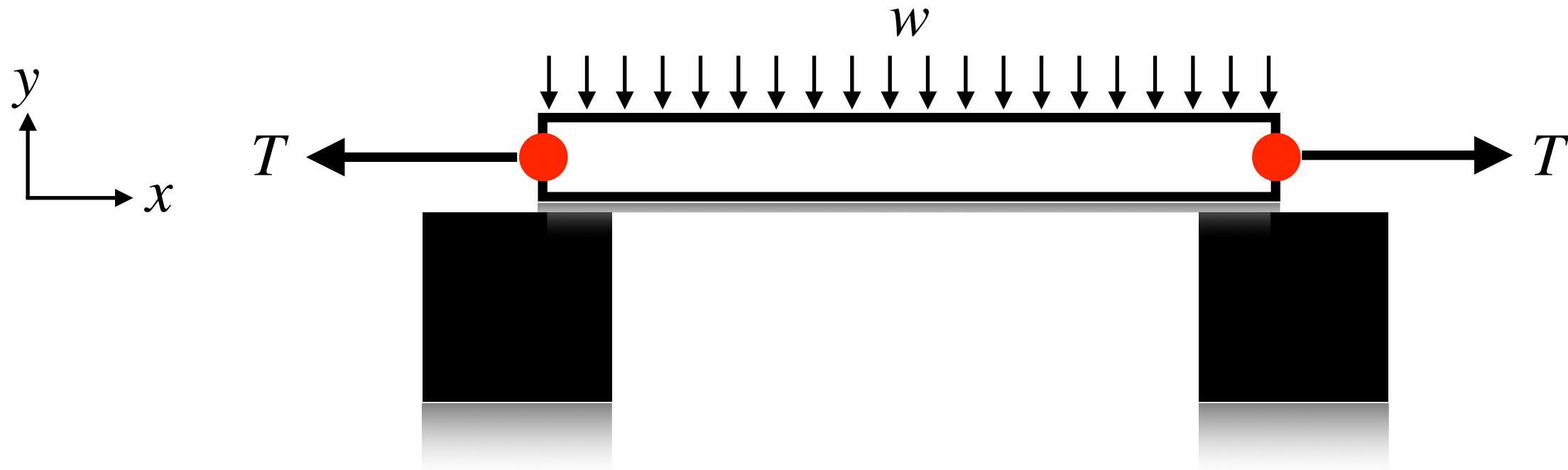
Neumann boundary condition.

Value of y given at the $x = b$:
Dirichlet boundary condition



Simple Beam Problem

A simply supported beam with a modulus of elasticity E , moment of inertia I , a uniform load w and end tension T is shown below.



The deflection at each point x in the beam can be calculated by solving the following differential equation:

$$y'' - ay = bx(L - x)$$

where L is the length of the beam and,

$$a = \frac{T}{EI}, \quad b = \frac{w}{2EI}$$

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = r(x)$$

$\parallel$
 0

$q(x) = -a$
 $r(x) = bx(L - x)$

The idea!

1. Approximate the derivatives by Finite Difference Approximation.
2. Implement boundary conditions
3. Solve the resulting set of algebraic equations.

$$\begin{aligned} y(a) &= \alpha \\ y(b) &= \beta \end{aligned} \quad \text{Dirichlet}$$

$$\begin{aligned} y'(a) &= \alpha && \text{Neumann at } x = a \text{ and} \\ y(b) &= \beta && \text{Dirichlet at } x = b \end{aligned}$$

$$[A]\{X\} = \{C\}$$

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = r(x)$$

$$\frac{d^2y}{dx^2} \approx \frac{y_{i-1} - 2y_i + y_{i+1}}{\Delta^2}$$

$$\frac{dy}{dx} \approx \frac{y_{i+1} - y_{i-1}}{2\Delta}$$

Boundary Value Problem: Finite Difference Approach

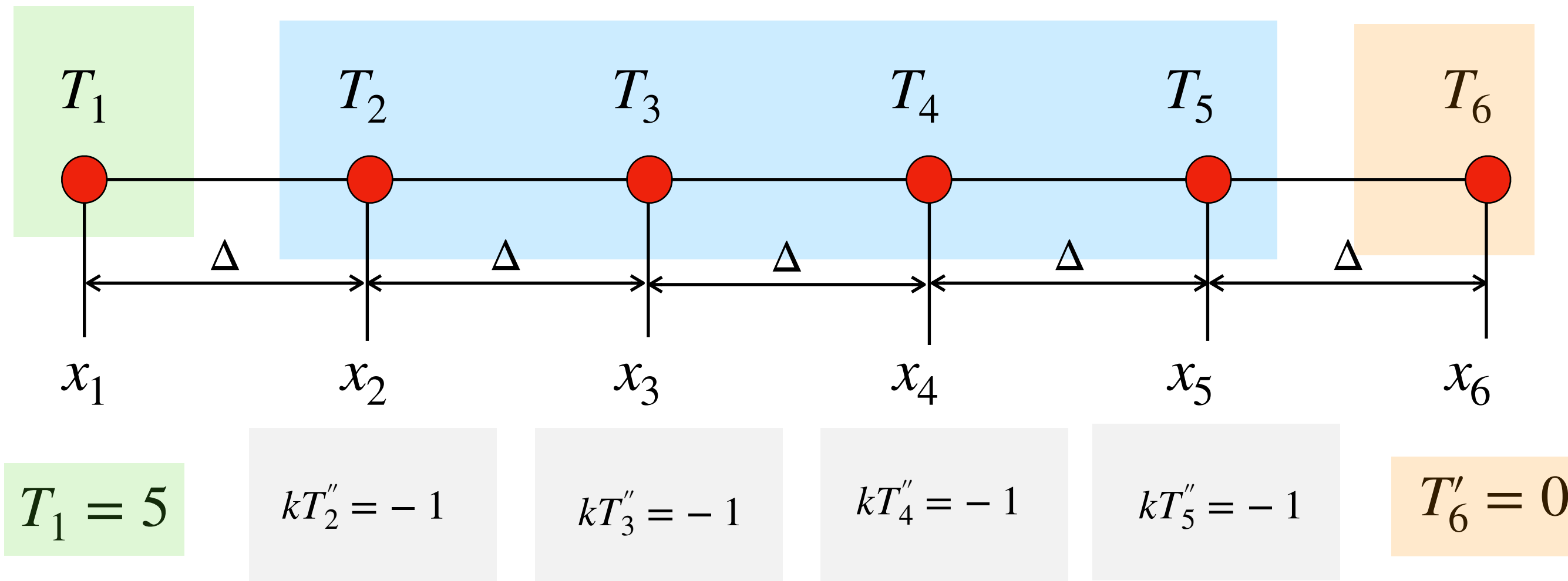
First consider the case where you “discretise” your domain $0 \leq x \leq 1$ into 5 intervals.

$$k \frac{d^2 T}{dx^2} = -1$$

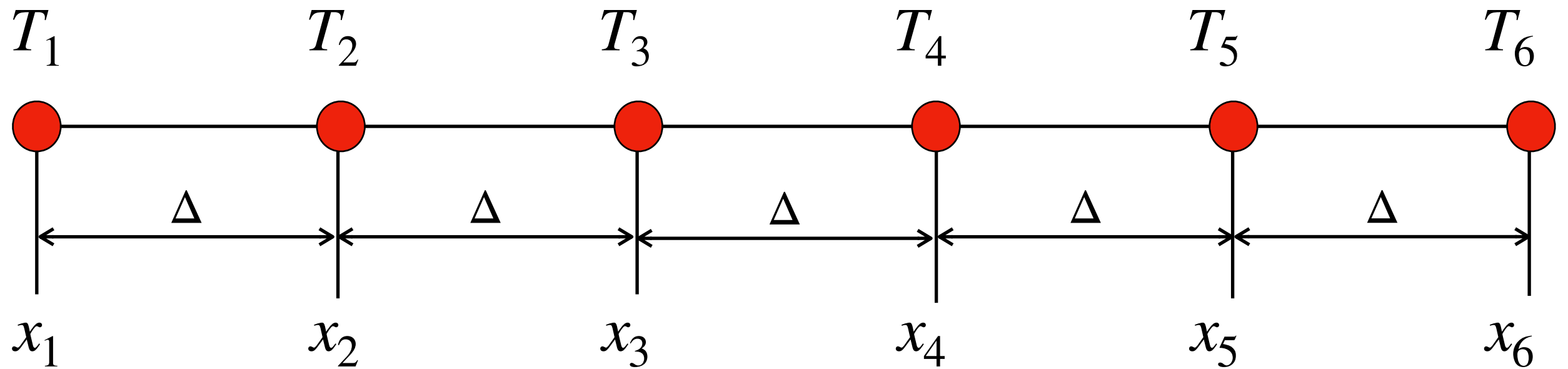
$$T_1 = 5$$

$$\left(\frac{dT}{dx}(1) = 0.0 \right)$$

Value of T given at $x = 0$:
Value of derivative given at $x = 1$.



So you need to solve for the unknowns T_2, T_3, T_4, T_5 and T_6 .



$$T_1 = 5$$

$$kT_2'' = -1$$

$$kT_3'' = -1$$

$$kT_4'' = -1$$

$$kT_5'' = -1$$

$$T_6' = 0$$

Using
Central difference
approximation

$$k \begin{Bmatrix} T_2'' \\ T_3'' \\ T_4'' \\ T_5'' \end{Bmatrix} = \begin{Bmatrix} -1 \\ -1 \\ -1 \\ -1 \end{Bmatrix}$$

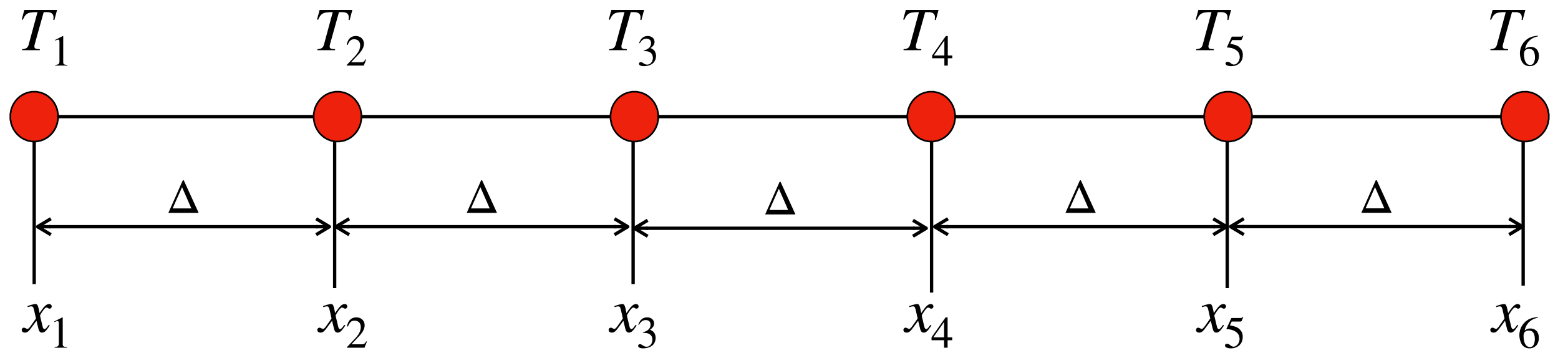
Using Central difference
approximation

$$T_2'' = \frac{T_3 - 2T_2 + T_1}{\Delta^2}$$

$$T_3'' = \frac{T_4 - 2T_3 + T_2}{\Delta^2}$$

$$T_4'' = \frac{T_5 - 2T_4 + T_3}{\Delta^2}$$

$$T_5'' = \frac{T_6 - 2T_5 + T_4}{\Delta^2}$$



$$T_1 = 5 \quad kT_2'' = -1 \quad kT_3'' = -1 \quad kT_4'' = -1 \quad kT_5'' = -1 \quad T_6' = 0$$

$$\begin{aligned}
 T_2'' &= \frac{T_3 - 2T_2 + T_1}{\Delta^2} \\
 T_3'' &= \frac{T_4 - 2T_3 + T_2}{\Delta^2} \\
 T_4'' &= \frac{T_5 - 2T_4 + T_3}{\Delta^2} \\
 T_5'' &= \frac{T_6 - 2T_5 + T_4}{\Delta^2}
 \end{aligned}
 \quad
 k \begin{Bmatrix} T_2'' \\ T_3'' \\ T_4'' \\ T_5'' \end{Bmatrix} = \begin{Bmatrix} -1 \\ -1 \\ -1 \\ -1 \end{Bmatrix}$$

$$\frac{1}{\Delta^2} \begin{bmatrix} 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{Bmatrix}$$

$$T_1 = 5$$

$$\frac{1}{\Delta^2} \begin{bmatrix} 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{Bmatrix} = \begin{Bmatrix} -1/k \\ -1/k \\ -1/k \\ -1/k \end{Bmatrix}$$

$$\begin{aligned} T'_6 &= 0 \\ (T_6 - T_5)/\Delta &= 0 \\ T_6 &= T_5 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 & 0 & 0 \\ 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 & 0 \\ 0 & 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 \\ 0 & 0 & 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{Bmatrix} = \begin{Bmatrix} 5 \\ -1/k \\ -1/k \\ -1/k \\ -1/k \\ 0 \end{Bmatrix}$$

This is now a 6x6 set of linear algebraic system of equations. You can solve to get

$$\begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{Bmatrix}$$

